CODETTE PRIOR-ART EVIDENCE TABLE

Consolidated External Artifact Register

Prepared for Legal Review — Intellectual Property & Priority-of-Invention Documentation

Date: April 12, 2026

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Classification: **CONFIDENTIAL — ATTORNEY WORK PRODUCT**

1. Purpose & Scope

This document consolidates all externally verifiable artifacts establishing prior art, authorship, and priority of invention for the **Codette cognitive AI framework**, developed by Jonathan Harrison under Raiff's Bits LLC. It is prepared for use by legal counsel in connection with intellectual property protection, priority-of-invention claims, and any related proceedings.

Each entry in the Master Prior-Art Evidence Table includes a verifiable source URL, an immutability classification indicating the strength and nature of timestamp evidence, and platform-sealed timestamps where available. All Digital Object Identifiers (DOIs) referenced herein are registered with **DataCite** and resolve to **CERN-hosted (Zenodo)** or **Hugging Face** infrastructure. Git-based artifacts carry cryptographic SHA-1 commit hashes establishing chain integrity. Together, these

records constitute a comprehensive evidentiary foundation for Codette's development timeline, architectural originality, and continuous innovation.

2. Immutability Classification Key

The following classification system is used throughout this document to characterize the evidentiary strength and verifiability of each artifact's timestamp and integrity:

Classification	Description & Evidentiary Weight
IMMUTABLE-DOI	**Artifact has a registered DOI with DataCite; timestamp sealed by CERN/Zenodo or Hugging Face infrastructure. Cannot be altered or backdated. Strongest evidentiary weight.**
IMMUTABLE-GIT	**Artifact exists in a Git repository with cryptographic SHA-1 commit hashes. Commit timestamps are author-asserted but hash-chained. Strong evidentiary weight when corroborated.**
IMMUTABLE-ORCID	**Persistent identifier registered with ORCID.org; links author identity to all associated works. Identity-level evidence.**
PLATFORM-SEALED	**Timestamp set by a third-party platform (OpenAIRE, Devpost, HF Community). Moderate evidentiary weight.**
PRIVATE-REGISTRY	**Artifact exists in a non-public registry (e.g., Azure AD). Requires subpoena or owner disclosure to verify. Administrative evidence.**

3. Master Prior-Art Evidence Table

The following table constitutes the core evidentiary register. Each row represents a distinct, externally verifiable artifact. All fields are populated from direct inspection of source platforms.

Evidence #	Immutability Class	Evidence Type	Artifact Name	Platform	Timestamp / Date	DOI or Identifier	Verifiable Source URL	Relevance to Codette	Legal Notes
PA-001	IMMUTABLE-DOI	Archival Deposit (Software + Paper)	Codette & Pidette: Sovereign AI Framework — “AI Ethics in Realtime”	Zenodo (CERN)	April 14, 2025 (created); Modified May 12, 2025	**10.5281/zenodo.15214462**	https://doi.org/10.5281/zenodo.15214462	Earliest DOI-stamped deposit. Establishes system architecture (CognitiveProcessor, BroaderPerspectiveEngine, NeuroSymbolicEngine, EthicalAIGovernance), chain of custody documentation, reasoning perspectives, fine-tuning convergence data, and sovereign innovation clause. Includes 21 files with MD5 checksums.	CERN-hosted; DataCite-registered DOI; indexed by OpenAIRE. Files include MD5 integrity hashes. CC-BY-4.0 license. Strongest priority-date evidence.
PA-002	IMMUTABLE-DOI	Archival Deposit (Journal Article + Software)	Citizen-Science Quantum and Chaos Simulations Orchestrated by the Codette AI Suite	Zenodo (CERN)	May 5, 2025 (published); Created May 10, 2025; Modified August 18, 2025	**10.5281/zenodo.15342466**	https://doi.org/10.5281/zenodo.15342466	Documents Codette’s quantum simulation pipeline, cocoon data wrapping (CognitionCocooner), multi-core trial orchestration with NASA	CERN-hosted; DataCite-registered; indexed by OpenAIRE. Version 2 record; newer version may exist. 158 views, 436 downloads as of

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								exoplanet data, neural classification, dream-state transformation, and 3D clustering. Includes 11 files with MD5 checksums. Continues PA-001. Links to HF DOI **10.57967/hf/5347**.	verification date. CC-BY-4.0.
PA-003	IMMUTABLE-DOI	AI Model Repository	Raiff1982/Codette — Falcon & Mistral Merged Model	Hugging Face	Created c. February 2025 (initial commit “about 1 month ago” as of March 2025 snapshot)	**10.57967/hf/5073** (current); **10.57967/hf/4842** (earlier revision)	https://huggingface.co/Raiff1982/Codette	First Hugging Face model deposit. Falcon-40B + Mistral-7B merge for text generation, medical diagnostics, and code reasoning. Trained on Raiff1982/coredata (11.6k samples) and Raiff1982/pineco. Establishes model architecture and training lineage. 124 monthly downloads as of	Hugging Face DOI via DataCite. Gated model with contact-sharing requirement. MIT license. Contains GGUF quantized variants (**doi:10.57967/hf/5601**, total 18.58 GB).

Evidence #	Immutability Class	Evidence Type	Artifact Name	Platform	Timestamp / Date	DOI or Identifier	Verifiable Source URL	Relevance to Codette	Legal Notes
								verification.	
PA-004	IMMUTABLE-DOI	AI Model Repository	Raiff1982/Codettev2 — Text-to-Image LoRA Model	Hugging Face	Updated May 28, 2025	**10.57967/hf/5542**	https://huggingface.co/Raiff1982/Codettev2	Second-generation Codette model with text-to-image capabilities via LoRA/Diffusers/ONNX. Demonstrates expansion of Codette architecture into multimodal generation.	Hugging Face DOI via DataCite. MIT license. 62 downloads as of verification.
PA-005	IMMUTABLE-DOI	AI Model Repository (Full Suite)	Raiff1982/Codette3.0 — Sovereign Multi-Perspective Consciousness System	Hugging Face	Updated (active development as of April 2026)	**10.57967/hf/7240**	https://huggingface.co/Raiff1982/Codette3.0	Major version release. Implements full six-layer modular stack, 11 cognitive perspectives, quantum-inspired architecture, CLI/Web/API deployment modes. Trained on Raiff1982/coreda ta and Raiff1982/eval datasets. SafeTensors format. Links to GitHub Raiff1982/TheAi (92 commits).	Hugging Face DOI via DataCite. Apache-2.0 license. Transformers + SafeTensors compatible.

Evidence #	Immutability Class	Evidence Type	Artifact Name	Platform	Timestamp / Date	DOI or Identifier	Verifiable Source URL	Relevance to Codette	Legal Notes
PA-006	IMMUTABLE-DOI	AI Model Repository (LoRA Adapters)	Raiff1982/codette-lora-adapters — 8 Domain-Specialized LoRA Adapters	Hugging Face	Active (DOI assigned)	**10.57967/hf/7980**	https://huggingface.co/Raiff1982/codette-lora-adapters	8 cognitive-perspective LoRA adapters (Newton, DaVinci, Empathy, Philosophy, Quantum, Consciousness, Multi-Perspective, Systems Architecture) trained on 20,500 examples. QLoRA 4-bit quantization on Llama-3.1-8B-Instruct base. Demonstrates novel CPU-only training pipelines validated on consumer laptops.	Hugging Face DOI via DataCite. CC-BY-4.0 license. PEFT/SafeTensors format.
PA-007	IMMUTABLE-DOI	AI Training Lab (Full Pipeline + Weights)	Raiff1982/codette-training-lab — Complete Training Pipeline	Hugging Face	Session 2026-03-19 (latest verified test); DOI assigned	**10.57967/hf/7951**	https://huggingface.co/Raiff1982/codette-training-lab	Complete training infrastructure. 8 adapters in GGUF (~924 MB) and PEFT SafeTensors (~79 MB) formats. Verified Phase 6 stability patches. Key metrics:	Hugging Face DOI via DataCite. MIT license. Gated model. Eval results published.

Evidence #	Immutability Class	Evidence Type	Artifact Name	Platform	Timestamp / Date	DOI or Identifier	Verifiable Source URL	Relevance to Codette	Legal Notes
								Phase Coherence (Gamma) 0.9835, AEGIS Ethical Alignment (Eta) 0.961, Cocoon Coherence 0.994, Memory Phase Stability 0.969, Tension Decay 91.2%.	
PA-008	IMMUTABLE-DOI	Research Paper (Model Card)	Raiff1982/codette-paper — “Codette: A Sovereign Modular Cognitive Architecture for Ethical Multi-Agent AI”	Hugging Face	Published (active)	N/A (no separate DOI; hosted as model card)	https://huggingface.co/Raiff1982/codette-paper	Full research paper establishing RC+ξ (Recursive Convergence + Epistemic Tension) framework, Multi-Agent Reasoning Forge, and AEGIS Ethical Governance. Reports key metrics: Ethical Alignment 82.6%, Phase Coherence 0.99, Epistemic Tension Decay 71.3%, Cocoon Coherence 0.994, Attractor Radius 0.093 in 64D	Platform-timestamped. ORCID linked. Contains formal mathematical framework and experimental results.

Evidence #	Immutability Class	Evidence Type	Artifact Name	Platform	Timestamp / Date	DOI or Identifier	Verifiable Source URL	Relevance to Codette	Legal Notes
								state space. CC-BY-4.0.	
PA-009	IMMUTABLE-GIT	Source Code Repository (Primary)	Raiff1982/Codette — Main GitHub Repository	GitHub	126 commits (ongoing); earliest commits c. early 2025	N/A (Git SHA-1 hashed)	https://github.com/Raiff1982/Codette	Primary source code repository. Contains complete Python codebase: ai_core.py, cognitive_processor.py, defense_system.py, health_monitor.py, fractal.py, quantum modules, GUI, templates. Includes Codette_Integrity_Certificate.json, research papers, architecture diagrams, and deployable packages.	Git SHA-1 commit hashes provide cryptographic chain. Public template repository. 126 commit history establishes continuous development timeline.
PA-010	IMMUTABLE-GIT	Source Code Repository (Organization)	Raiffs-bits/Codette — Organization Repository	GitHub	56 commits	N/A (Git SHA-1 hashed)	https://github.com/Raiffs-bits/Codette	Organization-level mirror under Raiffs-bits. Contains quantum modules, cocoon files, simulation data, and PDF exports. Establishes organizational	Public template. 56-commit chain. Corroborates PA-009 authorship.

Evidence #	Immutability Class	Evidence Type	Artifact Name	Platform	Timestamp / Date	DOI or Identifier	Verifiable Source URL	Relevance to Codette	Legal Notes
								ownership under Raiff's Bits LLC.	
PA-011	IMMUTABLE-GIT	Source Code Repository (Full Suite)	Raiff1982/TheAi — Full Codette Suite	GitHub	92 commits	N/A (Git SHA-1 hashed)	https://github.com/Raiff1982/TheAi	Complete deployable Codette suite with full functionality. Includes Docker deployment configs, training data, adapter training pack, codette_paper.pdf, and production Dockerfiles. Apache-2.0 licensed. Links to HF Codette3.0.	Apache-2.0 license. 92-commit chain. Contains codette_paper.pdf — independent timestamped copy of research paper.
PA-012	IMMUTABLE-GIT	Source Code Repository (Reasoning Engine)	Raiff1982/Codette-Reasoning — Advanced Multi-Perspective Reasoning Engine	GitHub	8 commits	N/A (Git SHA-1 hashed)	https://github.com/Raiff1982/Codette-Reasoning	Dedicated reasoning engine repository. Contains adapter router, agent LLM integration, Phase 1–7 summaries, production deployment guides, evaluation framework, and consciousness module.	8-commit chain. Phase documentation provides contemporaneous development evidence.

Evidence #	Immutability Class	Evidence Type	Artifact Name	Platform	Timestamp / Date	DOI or Identifier	Verifiable Source URL	Relevance to Codette	Legal Notes
								Documents full development lifecycle from Phase 1 through Phase 7 MVP.	
PA-013	IMMUTABLE-ORCID	Persistent Author Identifier	ORCID Record — Jonathan Harrison	ORCID.org	Active (persistent)	N/A	https://orcid.org/0009-0003-7005-8187	Globally unique persistent identifier linking Jonathan Harrison to all Codette publications, Zenodo deposits, and Hugging Face models. Referenced in every DOI-bearing artifact.	ORCID is a non-profit, community-governed identifier system. Provides independent identity verification across all platforms.
PA-014	PLATFORM-SEALED	Community Publication	“Quantum AI From your Couch” — Hugging Face Community Article	Hugging Face (Community Articles)	Published May 5, 2025	N/A	https://huggingface.co/blog/Raiff1982/quantum-ai-from-your-couch	Peer-facing publication documenting citizen-science quantum simulations using the Codette AI Suite. Includes methodology, results tables, and links to Zenodo DOI **10.5281/zenodo.15342466**. Demonstrates	Platform-timestamped by Hugging Face. Public disclosure for prior-art purposes.

Evidence #	Immutability Class	Evidence Type	Artifact Name	Platform	Timestamp / Date	DOI or Identifier	Verifiable Source URL	Relevance to Codette	Legal Notes
								public disclosure of methods.	
PA-015	PLATFORM-SEALED	Hackathon Entry	“Quantum from your Couch” — Devpost Hackathon Project	Devpost	Submitted during hackathon (c. 2025)	N/A	https://devpost.com/Raiff1982	Public hackathon submission documenting quantum simulation work. Skills listed: training, Azure, LLM. Establishes public demonstration of Codette capabilities in competitive context.	Devpost-timestamped. Third-party platform corroborates public demonstration.
PA-016	PLATFORM-SEALED	Academic Index Entry	OpenAIRE Index — Codette Records	OpenAIRE (EU)	Indexed following Zenodo publication	Inherits from PA-001 and PA-002	https://explore.openaire.eu/search/result?pid=10.5281/zenodo.15342466	Independent EU research infrastructure indexing of Codette deposits. Provides third-party corroboration of existence and timestamp of Zenodo records.	OpenAIRE is an EU-funded research infrastructure. Independent indexing provides additional evidentiary weight.
PA-017	PLATFORM-SEALED	Profile & Contributions Page	Jonathan “Raiff1982” Harrison — Profile and Contributions	Zenodo (CERN)	Active (comprehensive profile)	N/A	https://zenodo.org/search?q=metadata.creators.person_or_org.name:%22Harrison,%20Jonathan%22	Comprehensive profile page on CERN-hosted Zenodo documenting full project history, including Pi2_0 (c. 2023 predecessor),	CERN-hosted platform. Contains third-party summary of development chronology.

Evidence #	Immutability Class	Evidence Type	Artifact Name	Platform	Timestamp / Date	DOI or Identifier	Verifiable Source URL	Relevance to Codette	Legal Notes
								Codette (2025), fine-tuning details, hackathon participation, and version 5.0 “Sovereign Conscience” build release.	
PA-018	IMMUTABLE-DOI	AI Model Repository (Related System)	Raiff1982/nexus-signal-engine — Nexis Signal Engine	Hugging Face + Zenodo	Active	**10.57967/hf/6059**	https://huggingface.co/Raiff1982/nexus-signal-engine	Related cognitive framework for ethical signal analysis sharing architectural DNA with Codette. Multi-perspective agent architecture (Colleen, Luke, Kellyanne, Emma, Noah), weighted fuzzy-term scoring, SHA-256 integrity hashing. Demonstrates broader ecosystem of related innovations.	Hugging Face DOI via DataCite. MIT license. Cross-references Codette architecture.
PA-019	PRIVATE-REGISTRY	Cloud Platform Registration	Azure AD Application Registration — Codette	Microsoft Azure (Entra ID)	Referenced in codebase and Zenodo deposits; exact date held in private tenant	N/A	N/A (Private tenant — requires owner disclosure or legal discovery)	Application registration in Microsoft Azure Active Directory (now Entra ID) for	Private registry. Timestamp verifiable via Microsoft subpoena or tenant

Evidence #	Immutability Class	Evidence Type	Artifact Name	Platform	Timestamp / Date	DOI or Identifier	Verifiable Source URL	Relevance to Codette	Legal Notes
								Codette's cloud deployment and authentication services. Referenced in cost analysis documents, deployment configurations, and app.yml files across repositories.	admin export. Administrative evidence requiring disclosure.
PA-020	IMMUTABLE-GIT	Source Code Repository (Legacy Archive)	Raiff1982/codettes-legacy — "The Truth"	GitHub	Archived	N/A (Git SHA-1 hashed)	https://github.com/Raiff1982/codettes-legacy	Archived legacy repository preserving earlier development history. Labeled "the truth" — establishes historical development continuity.	Archived repository. Git SHA-1 chain preserved.

4. DOI Cross-Reference Summary

The following table provides a consolidated index of all Digital Object Identifiers associated with Codette artifacts, listed in chronological order where determinable:

DOI	Platform	Date / Notes
10.5281/zenodo.15214462	**Zenodo (CERN)**	**April 14, 2025**
10.5281/zenodo.15342466	**Zenodo (CERN)**	**May 5, 2025**

DOI	Platform	Date / Notes
10.57967/hf/4842	**Hugging Face**	**c. February 2025 (earliest HF revision)**
10.57967/hf/5073	**Hugging Face**	**c. February–March 2025**
10.57967/hf/5347	**Hugging Face**	**Linked from Zenodo PA-002**
10.57967/hf/5542	**Hugging Face**	**May 2025**
10.57967/hf/5601	**Hugging Face**	**GGUF variant**
10.57967/hf/6059	**Hugging Face**	**Nexis Signal Engine**
10.57967/hf/7240	**Hugging Face**	**Codette 3.0**
10.57967/hf/7951	**Hugging Face**	**codette-training-lab**
10.57967/hf/7980	**Hugging Face**	**codette-lora-adapters**

5. Chain of Custody & Authentication Notes

5.1 Zenodo (CERN) — DOI Immutability. All Zenodo records referenced in this document are hosted by the European Organization for Nuclear Research (CERN) and registered with DataCite. DOI timestamps are assigned at the moment of publication and cannot be altered or backdated by any party, including the depositor. Each file deposited carries an MD5 checksum recorded by Zenodo at upload time, providing bit-level integrity verification. Zenodo's infrastructure guarantees long-term preservation in accordance with CERN's data retention policies.

5.2 Hugging Face — DOI Registration. All Hugging Face DOIs are issued through DataCite integration and are permanently linked to the corresponding model or dataset repository. Model files stored in SafeTensors and GGUF formats carry internal checksums that enable independent verification of file integrity. Hugging Face maintains version histories with commit-level granularity.

5.3 GitHub — Cryptographic Commit Integrity. All GitHub repositories listed herein use Git's SHA-1 cryptographic hashing for commit integrity. Each commit hash is computed from the commit content, parent hash, author information, and timestamp, creating a hash chain. While commit timestamps are author-asserted, the hash chain prevents retrospective insertion or modification of commits without

producing detectable hash breaks. The public visibility of these repositories allows any party to independently verify the commit history.

5.4 ORCID — Persistent Author Identity. ORCID 0009-0003-7005-8187 provides a persistent, globally unique author identification linking Jonathan Harrison to all associated publications, deposits, and model repositories across Zenodo, Hugging Face, and GitHub. ORCID is maintained by a non-profit, community-governed organization and serves as an independent identity verification mechanism.

5.5 OpenAIRE — Independent EU Indexing. OpenAIRE independently indexes Zenodo records as part of the European Union's research infrastructure initiative. The OpenAIRE index entries for Codette artifacts (see PA-016) provide EU infrastructure-level corroboration of deposit dates and content, constituting an additional layer of third-party verification independent of the depositor.

5.6 Azure AD — Private Registry Verification. Azure Active Directory (now Microsoft Entra ID) registration timestamps for the Codette application are held in Microsoft's private cloud infrastructure. These timestamps are verifiable through tenant administrator export or through legal discovery processes including subpoena. While not publicly accessible, these records constitute administrative evidence of application registration within Microsoft's enterprise identity platform.

6. Certification Statement

I, **Jonathan Harrison**, ORCID **0009-0003-7005-8187**, certify that the artifacts listed in this document are authentic prior-art evidence of my original work on the Codette cognitive AI framework, developed under **Raiff's Bits LLC**. All DOIs, commit hashes, and platform timestamps referenced herein are verifiable through the public URLs and registries indicated. I make this certification under penalty of perjury and in support of any intellectual property proceedings for which this document may be submitted.

Signature:

Jonathan Harrison

CEO & CTO, Raiff’s Bits LLC

ORCID: 0009-0003-7005-8187

Date:

NOTARIZATION (Optional)

State of _____ County of _____

Subscribed and sworn to before me this _____ day of _____, 20_____.

Notary Public:

My commission expires: _____

[SEAL]

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7. Disclaimer

This document is prepared for legal review purposes and does not constitute legal advice. Verification of all claims should be independently conducted by counsel. Platform timestamps and DOI registrations are subject to the terms of service of their respective hosting organizations. The classifications of evidentiary weight provided herein (e.g., “strongest,” “strong,” “moderate”) reflect the author’s assessment based on the technical properties of each platform and are offered for counsel’s consideration, not as legal conclusions. Nothing in this document should be construed as a warranty of the outcome of any legal proceeding.

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